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Drivers' recognition of pedestrian road-crossing intentions: Performance and process

Wenxiang Chen, Xiangling Zhuang *, Zixin Cui, Guojie Ma *

Shaanxi Key Laboratory of Behavior and Cognitive Neuroscience, School of Psychology, Shaanxi Normal University, Xi'an, China

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ABSTRACT

Drivers' recognition of pedestrian road crossing intentions is an essential process during driver-pedestrian interaction. However, compared with the rich observational findings on interaction behavior, little is known on drivers' performance in recognizing pedestrian intentions, as well as the underlying cognitive processes. To fill in the gap, this study evaluated drivers' performance in making judgments of pedestrians' road crossing intentions in recorded natural driving scenes. Experienced and novice drivers identified pedestrians as "will cross" or "will not cross" at some time-to-arrival while their eye movements were recorded. The results showed that experienced drivers were more conservative in discriminating whether a pedestrian would cross or not (preferred a "pedestrian will cross" judgment) and took a higher level of information processing of pedestrian intention. Regardless of driving experience, drivers had a higher detection rate, earlier detection, higher level of information processing and quicker response over pedestrians who intended to cross than those did not intend to cross. A quicker response was also achieved when the time-to-arrival was smaller. Analysis of eye movements showed attentional bias to the upper body of pedestrians when recognizing intention. These findings offer an initial understanding of the intention recognition process during driver-pedestrian interaction and inform directions for autonomous driving research when interacting with pedestrians.

1. Introduction

Pedestrians are the most vulnerable road users that accounting for 22% of the total fatalities in traffic accidents (World Health Organization, 2013). The issue of pedestrian safety is more serious in low- and middle-income countries (Naci, Chisholm, & Baker, 2009). Since pedestrian accidents usually occur as a result of unsafe interaction with vehicles, it is critical to ensure safe interaction to increase pedestrian safety. To that end, researchers have applied several methods including accident analysis (Jacobsen, 2003; Murphy, Levinson, & Owen, 2017), conflicts identification and categorization (Ni, Wang, Sun, & Li, 2016), and behavioral observation (Kaparias, Bell, Biagioli, Bellezza, & Mount, 2015) to gain a deeper understanding of the nature of the driver-pedestrian interaction.

Among these approaches, behavioral observation targets at the microscopic interaction behaviors instead of only analyzing the final consequences of interaction (e.g., accidents or conflicts). For example, it may concern whether a driver will yield to pedestrians and how to promote the yielding behavior (Zhuang & Wu, 2014). However, the observed behaviors of both drivers and pedestrians are by no means the fundamental causes of accidents, as at an upper level of analysis, the behaviors

* Corresponding author.
E-mail addresses: zhuangxl@snnu.edu.cn (X. Zhuang), magj@snnu.edu.cn (G. Ma).

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themselves are the *consequence* of information processing during the interaction. According to the theory of situation awareness which comprises three levels (Endsley, 1995; Gugerty, 2011), drivers must detect pedestrians, recognize the intentions of pedestrians, and predict the future state of pedestrians before making a behavioral response. Therefore, ensuring safe interactions requires a full understanding of information processing at all three levels. Nevertheless, our review shows that although evidence on the first level of situation awareness (i.e., how drivers detect pedestrians) has accumulated for decades (for a review, see Langham & Moberly, 2003; Tyrrell, Wood, Owens, Whetsel Borzendowski, & Stafford Sewall, 2016), studies on the second level of situation awareness (i.e., how drivers recognize pedestrian intention) are still rare. Therefore, in the following section, we first inferred the role of intention recognition in the driver-pedestrian interaction through drivers' responses to pedestrians' communication behaviors. After that, we reviewed a few scarce studies on driver's intention recognition process in Section 1.2 and proposed our research question in Section 1.3.

1.1. The role of intention recognition in driver response to pedestrians

We have mentioned that driver responses to pedestrians are only the consequence of information processing during the driver-pedestrian interaction. That said, behavioral responses still offer some clues on how drivers process information about pedestrian intentions. Previous studies showed that drivers were more likely to yield to pedestrians if pedestrians actively took some actions like displaying hand gestures (Crowley-Koch, Van Houten, & Lim, 2011; Zhuang & Wu, 2014) to convey their intentions to drivers. Guéguen, Eyssartier, and Meineri (2016) examined the effect of the pedestrians' face expressions on drivers' yielding behavior and found increased yielding behavior for pedestrians who smiled at the approaching vehicle. Similar to the effect of the smile, even simple eye contact can increase yielding behavior (Guéguen, Meineri, & Eyssartier, 2015) or lead drivers to start decelerating at a longer distance with a smoother process (Ren, Jiang, & Wang, 2016). Besides the effect of explicit communication cues, the yielding rate also showed a strong correlation with the natural crossing behavior of pedestrians (Lu, Ren, Wang, Chan, & Wang, 2016; Schroeder & Rouphail, 2011). Several researchers suggested that drivers were more likely to yield if pedestrians displayed more assertive behaviors like walking fast to the crosswalk (Jiang, Wang, Mao, Bengler, & Bubb, 2011; Schroeder & Rouphail, 2011); stepping onto the crosswalk rather than waiting on the sidewalk (Harrell, 1993); leaving a shorter lateral distance (Jiang et al., 2011) or not paying attention to the traffic (Katz, Zaidel, & Elgrishi, 1975).

Why do drivers yield or show other cautious behaviors to pedestrians who display a specific behavior? For *explicit* communication methods like hand gesture, smile, eye contact, it is possible that drivers understand pedestrians' crossing intentions conveyed by them, thus decide to yield to them. Consistent with this conjecture, Dey and Terken (2017) classified the assertive behaviors of pedestrians (walk fast, step onto the roadway, etc.) as *implicit* communication behaviors, which meant that although pedestrians did not deliberately display these behaviors to convey the intentions to cross road. And these assertive behaviors could still be perceived as cues to drivers. Theoretically, this is possible because humans always understand others' goals and intentions by interpreting others' behaviors, which can invoke brain activities closely related to mirror neurons and action-observation network (AON) (Brass, Schmitt, Spengler, & Gergely, 2007; Kilner, 2011). Therefore, from the current findings, we can infer that drivers may rely on both explicit and implicit communication cues to recognize pedestrian intentions, which may be a general mechanism underlain all the factors that induced yielding behavior.

1.2. Drivers' recognition of pedestrian intention during interaction

The use of explicit and implicit cues to recognize intentions has been explored in a few studies involving road users beyond driver-pedestrian interaction. During the driver-driver interaction, intentions are usually recognized by formal signals (like steering light, honking) and sometimes, when traffic regulations are ambiguous in some specific situations, drivers use informal signals (like hand gesture, approaching speed, etc.) to recognize intentions (Kitazaki & Myhre, 2015; Lee & Sheppard, 2016). For cyclists, they usually interact with drivers at junctions where the most driver-cyclist collisions happened (Bonham & Johnson, 2018). Researchers have found that hand gesture, head movement, and speed information of cyclists helped drivers to recognize their intentions to turn left/right or go straight (Hemeren, Johannesson, Lebram, Eriksson, Ekman, & Veto, 2014; Walker, 2005; Westerhuis & De Waard, 2017).

For the driver-pedestrian interaction, there are two pioneering works focused on the intention recognition process. In the first study, drivers were asked to consider whether the pedestrian would cross the street and report which cues they utilized to make judgments (Schmidt, Färber, & Grassi, 2008, see in Schmidt & Färber, 2009). Following this, in the next study, participants were asked to make intention judgments when some of the cues in video clips were masked (Schmidt & Färber, 2009). The results showed that there might be five main cues utilized to recognize pedestrian intentions: head movement, legs movement, dynamics of walking, pedestrian characteristics and traffic environment. Since all these cues were not deliberately displayed by pedestrians, the finding suggested that drivers did use implicit communication cues to recognize pedestrian intentions.

While these two studies improved our understanding of drivers' intention recognition strategy, literature around intention recognition in traffic is still scarce. As a result, some basic questions on intention recognition remain unclear. Can drivers recognize pedestrian intentions in all contexts? If not, what factors may modulate their performance? Without answering these basic questions, even these two findings on intention recognition cues cannot be properly interpreted. For example, in Schmidt and Färber's study (2009), the videos were recorded in a third-party view without self-motions, and the performance was evaluated without time constraints. In contrast, in a normal driving scenario, drivers interact with several potential hazards. They need to detect potential risky pedestrians and recognize their intentions within a short duration. Besides, the moving of vehicle and the distance between drivers and pedestrians also affect the availability of some behavioral cues (e.g., head movements). It is unclear how these settings in natural driving influence intention recognition.

1.3. Objectives

As outlined, drivers need to recognize pedestrian intentions correctly and timely for safe interactions. However, only a few studies have explored the intention recognition process. We have gleaned some indirect evidence from drivers' responses to pedestrian behaviors to infer its role in driver yielding behavior, but the inference is just theoretical speculation. The two relevant studies (Schmidt et al., 2008; Schmidt & Färber, 2009) offered valuable findings on drivers' strategies in intention recognition, but the experiment settings neglected the viewpoint of drivers, the distance to pedestrians, as well as the time limit of intention recognition that was crucial in realistic driving scenarios. Overall, the field of intention recognition is still in its infancy.

In this study, we evaluated drivers' performance and process in intention recognition and how it was modulated by some important factors. To overcome potential limitations in previous research, we simulated the task from the detection of pedestrians to the recognition of their intentions in drivers' viewpoint while scenes were moving towards drivers. The distance to pedestrians was reflected by the time gap between drivers and pedestrians (Time-to-arrival, TTA) at the moment of requiring drivers to make an intention judgment. What's more, previous researches on action understanding has shown that observer's expectation acquired from experience also played a role in intention recognition beyond sensory information conveyed by movement kinematics (Chambon et al., 2017, 2011). Applying the findings into the current study, driving experience is likely to affect performance in intention judgments. Therefore, we included both experienced and novice drivers as participants to make intention judgments toward pedestrians who had the intention to cross or not. The recorded eye movements and reaction time in judgments helped to identify the process as well as the cues of intention recognition in the driver-pedestrian interaction. Overall, the current study evaluated drivers' performance in intention judgments, and how it was affected by drivers' experience, distance to pedestrians (in fact, TTAs), and real intentions of pedestrians.

2. Methods

2.1. Design

A 2 (driving experience) $\times$ 2 (pedestrian intentions) $\times$ 4 (TTAs) mixed design was employed. The between-subject factor was driving experience (experienced drivers versus novice drivers). The within-subject factors were intentions of pedestrians (cross or non-cross) and TTAs (2, 3, 4, 5 s before the vehicle arrived at the intersection of vehicle and pedestrian's paths). The dependent variables were the performance of drivers in intention judgments, including accuracy, response bias, speed of responses, and eye movements measures.

2.2. Participants

A total of 40 full-licensed drivers (29 males and 11 females) were paid to take part in the experiment. Twenty participants who had been driving for less than 10,000 km (range from 50 to 10, 000 km, $M = 1,031$ km, $SD = 2,369$ km) were classified as novice drivers. The others who had more than 100,000 km driving experience (range from 100,000 to 1500,000 km, $M = 431,500$ km, $SD = 337, 99$ km) were classified as experienced drivers. Novice drivers were undergraduate students recruited from the college and had an average age of 21.5 (SD = 1.93) with mean driving experience of 1.7 years. The experienced were professional drivers (e.g. taxi drivers) recruited by phone call for participation. Experienced drivers had an average age of 35.7 (SD = 7.4) with mean driving experience of 11.2 years. All participants had normal or corrected to normal visions.

2.3. Video clips used for simulation

A video-based simulation was provided to present traffic situations where drivers interacted with pedestrians. To offer a high-quality and natural simulation of interactions, a GoPro Hero 6 camera was mounted to the windscreen of vehicles to record traffic scenes from drivers' viewpoint. Five professional drivers were recruited for video recording in several days in Xi'an, a capital city of Shaanxi province in China where participants were recruited from. The drivers of recording vehicles were instructed to drive as usual and were naive to the purpose of the recording.

The raw videotapes, which had a 3840×2160 resolution at 60 Hz frame rate, were cut and compressed into a 3840×1080 resolution at 30 Hz frame rate, to play in the center of three monitors. The experimenter played the videotapes to select each driver-pedestrian interaction event based on some rules. First, during each interaction, the recording vehicle was the foremost vehicle in front of the pedestrian and did not decelerate, to ensure that the pedestrian intention was not a reaction to the yielding behavior of the recording driver or other vehicles. Second, the target pedestrian should be visible during the interaction process, and there were only one to three pedestrians crossing the road (not in a group of pedestrians). Third, there was no traffic light to assign the right of way. The procedure to generate video clips for the experiment is shown in Fig. 1.

First, we selected 117 interaction events, with eight of them for practice trials and 109 for the formal experiment. Here, "interaction event" was defined as the process from pedestrian appearing to the vehicle passing the pedestrian. The 109 events included 43 "cross" events where the pedestrian crossed before the recording vehicle, 57 "non-cross" events where the pedestrian did not cross (i.e., waited or slowed down), and 9 "noise" events where the pedestrian walked on the sidewalk. The noise events were included to promote a more active visual search of traffic scenes. All interactions took place in the urban environment and 16 of them took place when zebra crossing was present. The lanes of road ranged from 2 to 8 (2 lanes: 23 events; 3 lanes: 9 events; 4 lanes: 23 events; 6 lanes: 45 events; 8 lanes: 9 events).

Second, we cut interaction events into interaction clips with different TTAs, which is the time needed for the vehicle to arrive at the intersection between the vehicle and pedestrian's paths. For each interaction event, one to four clips were generated by stopping the driving scene at different TTAs (2, 3, 4, 5 s), which means that if TTA = 4 s, time between the final frame of the clip and the frame where vehicle arriving at the intersection is 4 s. In addition, if the pedestrian had stepped onto the lane or was obscured at the ending frame in one clip, this clip would be excluded for this interaction event. In the *Supplementary Materials*, four images according to four TTAs of one interaction event are provided as an example, as well as our video analysis of the clips.

Finally, we extended the interaction event to earlier normal driving phases of the video when pedestrians hadn't appeared in the scene (Step 3 in Fig. 1). The normal driving section provided a natural sense of interacting with pedestrians from the detection to responses and avoided abrupt appearing of pedestrians. The normal driving scene varied in length so that participants could not expect target pedestrians to appear at a fixed time. The final clips looked like this: A driver is driving normally, then pedestrians appeared in the scene. As a result, the length of composite clips displayed in the experiment varied between 7 and 15 s (2 s intervals).

2.4. Apparatus

The video clips were presented in the center of three 24-inch Dell U2414H monitors to simulate real traffic scenes. Each monitor had a 1920 × 1080 resolution. The angle between the contiguous screens was 135°. The center monitor was positioned facing the participant at a viewing distance of 62 cm. An EyeLink 1000 plus (SR Research Ltd, Ontario, Canada) eye tracker was used to track participants' eye movements at a sample rate of 1000 Hz in tower mode. Participants used a joystick to input their judgments of pedestrian intentions.

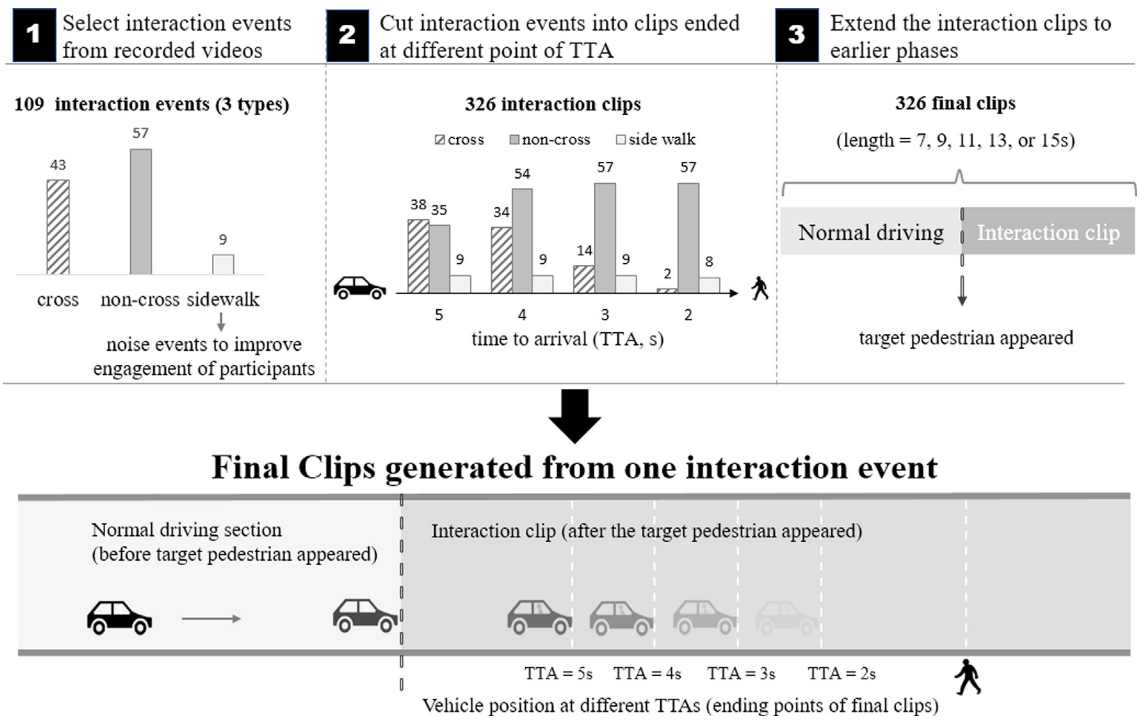

Fig. 1. The procedure to generate materials for the experiment. "Interaction clips" were generated from "interaction events". One "interaction event" could generate several clips according to TTAs, depending on whether the pedestrian had crossed the road and whether the pedestrian was visible at the end frame of the clip.

2.5. Procedure

The experiment was run in the same city as the video collection. The instruction and example picture (like Fig. 2) were shown on the center monitor. Participants were told that clips recorded from the drivers' viewpoint would be shown on monitors and they should view clips as if they were the driver and look around the traffic scenes as usual. When video stopped, they should judge pedestrian intentions (i.e., whether to cross in front of the approaching vehicle) and respond with a joystick (two keys assigned to two judgments).

Each participant first finished eight practice trials before the formal experiment. The 109 trials for the formal experiment were separated into three blocks included 36, 36 and 37 trials respectively to avoid fatigue. For each participant, the clip of each trial was randomly picked from a unique interaction event. Therefore, each participant watched an interaction event just once. A five-point calibration and validation procedure were used to calibrate the eye tracker at the beginning of each block on the center monitor.

For each trial, the first frame of the clip was displayed for 3 s as a preview of the traffic scene. After that, the clip started to play automatically. Immediately following the ending frame of the clip, a yellow rectangle (as shown in Fig. 2) surrounded the target pedestrian whose intention was to be judged. Both the rectangle and the last frame of the clip lasted for a maximum of 1 s. As instructed, participants needed to recognize pedestrian intentions (i.e., pedestrians would cross the road or not) and responded with a joystick as soon as possible. Finally, the trial would end with a black screen until participants responded (the maximum length of the black screen was 4 s). No feedback on whether pedestrians would cross the road was provided during the experiment.

The whole experiment lasted about 60 min. After the experiment on the computer, the participants also filled a questionnaire about driving experience and accident history. No accident history was reported.

3. Results

The recognition of pedestrian intention should be preceded by successful detection of pedestrians. In the following analyses, we first reported drivers' detection of pedestrians with eye tracking data. Then, in the main part, we evaluated drivers' performance in the judgment of pedestrian road crossing intentions from three aspects: accuracy, reaction time and eye movements. At last, we conducted an integrated analysis to explore the relationship between detection and reaction of intention recognition.

Notice that participants' responses for the condition where pedestrians just walking on the sidewalk were not included. And for clips with TTA of 2 s, pedestrians did not cross the road in most trials (690 in 710 trials, 97.1%), thus this level of TTA was also excluded for analysis employing ANOVA.

3.1. The detection of pedestrians during interaction

The dynamic area of interests (AOIs) in each frame surrounding the pedestrians were defined to evaluate drivers' detection of target pedestrians. The AOIs in each frame were about 1.5 times of the rectangle which closely surrounded a pedestrian (see the rectangle in Fig. 2 for an example). The wider AOI was used considering the accuracy of eye tracking and fixation locating, and the fact that pedestrians can be detected at a far distance when they only covered a small area of the display. In fact, the average size of AOIs only occupied around 1% (range from 0.1% to 6%) area of the center monitor. Eight percent of trials were excluded for having no fixation on the AOIs until the display turned black or the sample rate was lower than 80%, which might be caused by errors in eye tracking (e.g., participants moved the head, signal loss of eyes or inattention of participants).

Detection rate: If participants gazed at the target pedestrian before the presentation of rectangular indicator, this trial would be coded as "detected before indicator"; otherwise, it was coded as "not detected before indicator". There were 7.6% trials in which participants did not detect the pedestrian before indicator. A 2 (driving experience) × 2 (intention of

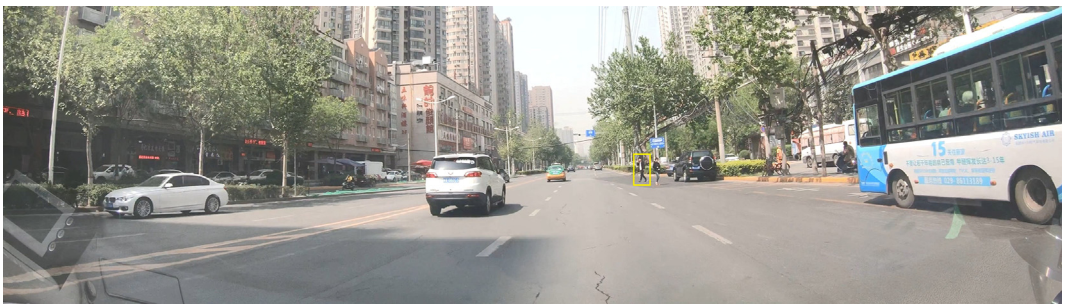

Fig. 2. For each trial, the rectangle indicates the target pedestrian to be judged. It appeared at the last frame of a clip and stay for one second.

pedestrian) × 3 (TTAs: 3, 4, 5 s) mixed ANOVA was employed to analyze the detection rates over groups and conditions. No significant effect of driving experience was found, $F < 1$. The main effects of the intentions and TTAs were significant ($F(1, 38) = 66.69$, $p < 0.001$, partial $\eta^2 = 0.637$; $F(2, 76) = 19.76$, $p < 0.001$, partial $\eta^2 = 0.342$). The detection rate was higher for pedestrians would cross (cross: $M = 0.95$, $SE = 0.006$; non-cross: $M = 0.86$, $SE = 0.011$), and for trials ended at shorter TTA (i.e., 3 s) (3 s: $M = 0.95$; 4 s: $M = 0.89$, 5 s: $M = 0.88$), all $ps < 0.001$. The interaction between intentions and TTAs was marginally significant, $F(2, 76) = 3.01$, $p = 0.055$, partial $\eta^2 = 0.073$. Post hoc tests showed that for each level of TTAs, the detection rate was always lower when pedestrians did not intend to cross than when pedestrians would cross, and the differences were greater for larger TTAs, all $ps < 0.002$.

Latency of detection and first fixation duration: The time taken to first fixate on target pedestrians (i.e., the latency of detection) was defined as the duration from the appearing of pedestrians to the moment that first fixation fell into AOIs. For the latency of detection and first fixation duration, a 2 (driving experience) × 2 (intention of pedestrian) mixed ANOVA was employed. The latency of detection was comparable for experienced and novice drivers, $F(1, 38) = 1.81$, $p = 0.186$, partial $\eta^2 = 0.046$, but varied with pedestrian intentions, $F(1, 38) = 5.71$, $p = 0.022$, partial $\eta^2 = 0.131$. More specifically, pedestrians intending to cross were detected earlier than those did not intend to cross (cross: $M = 1345$ ms, $SE = 57$ ms; non-cross: $M = 1445$ ms, $SE = 50$ ms), regardless of driver experience, $F < 1$.

For the first fixation duration, similar pattern of results was found. There was no group difference between experienced drivers and novices for the first fixation duration ($M = 634$ ms for experienced drivers and $M = 558$ ms for novices), $F(1, 38) = 1.69$, $p = 0.202$, partial $\eta^2 = 0.043$. All drivers had longer first fixation duration when pedestrians would cross ($M = 646$ ms, $SE = 39$ ms) than when pedestrians would not cross ($M = 545$ ms, $SE = 25$ ms), $F(1, 38) = 12.17$, $p = 0.001$, partial $\eta^2 = 0.243$. No interaction effect was found, $F < 1$.

3.2. Drivers' performance in judgments of pedestrian road crossing intentions

3.2.1. The accuracy, sensitivity and response bias in intention judgment

The ground truth of pedestrian intentions was defined as whether pedestrians crossed the road before the recording vehicle passing by. That is, if the pedestrians would cross the road in front of the recording vehicle at the end of this "interaction event", we would deem that the pedestrians crossed before the vehicle (*cross* events). Otherwise, the pedestrians would be regarded as not crossed before the vehicle, or in other words, the pedestrians may intend to cross the road but did not do so before the recording vehicle (*non-cross* events). It has to be mentioned that the rate of pedestrians did not cross the road decreased for larger TTAs (648 in 854 trials, 75.9% for 3 s; 602 in 1210 trials, 49.8% for 4 s; 340 in 1226 trials, 27.7% for 5 s).

Following the framework of signal detection theory (SDT), drivers' responses were classified into four categories. For *cross* events, the response was a hit (drivers made a "will cross" judgment) or miss (a "will not cross" judgment). For *non-cross* events, the response was a false alarm (a "will cross" judgment) or a correct rejection (a "will not cross" judgment). Hit rate was calculated as the number of hit responses divided by the number of cross trials; false alarm rate was calculated as the number of false alarm responses divided by the numbers of non-cross trials. The overall hit rates of experienced (0.92) and novice drivers (0.91) were equally high, $t = 0.26$, $p = 0.797$, but the false alarm rate of experienced drivers (0.43) was higher than novices (0.30), $t = 2.26$, $p = 0.029$.

Table 1 shows the hit rates and false alarm rates over driver groups and TTAs (3, 4, 5 s). Based on the hit rate and false alarm rate of each participant, the drivers' sensitivity (the overall ability to discriminate pedestrian intentions) and bias (the tendency to make "will cross" or "will not cross" judgments) were assessed according to SDT. Fig. 3 shows the sensitivity (d'), criterion (C) and response bias (β) over groups and TTAs. A higher d' score means the driver performed better in discriminating whether pedestrians would cross or not, whereas a lower C or higher β means the driver was more likely to make a "will cross" judgment. A 2 (driving experience) × 3 (TTAs: 3, 4, 5 s) mixed ANOVA was employed for analyses of parameters of SDT.

When sensitivity d' was regarded as the dependent variable, we found novice drivers ($M = 1.960$, $SE = 0.118$) performed better than experienced drivers ($M = 1.535$, $SE = 0.118$), $F(1, 38) = 6.49$, $p = 0.015$, partial $\eta^2 = 0.146$. The main effect of TTAs was also statistically significant, $F(2, 76) = 7.36$, $p = 0.001$, partial $\eta^2 = 0.162$. Pairwise comparisons confirmed that when the TTA was 4 s, the d' was higher than that at other TTAs (all $ps < 0.004$). The interaction effect was not statistically significant, $F(2, 76) = 0.21$, $p = 0.809$, partial $\eta^2 = 0.006$.

When criterion C was regarded as the dependent variable, we found the criterion of experienced drivers was marginally lower ($M = -0.625$, $SE = 0.086$) than that of novices ($M = -0.382$, $SE = 0.086$), $F(1, 38) = 4.02$, $p = 0.052$, partial $\eta^2 = 0.096$. The effect of TTAs and its interaction with driving experience were significant ($F(2, 76) = 10.24$, $p < 0.001$, partial $\eta^2 = 0.212$; F(2,

Table 1
Hit rates and false alarm rates over TTAs and groups.

	Hit rate (SE)			False alarm rate (SE)		
	3 s	4 s	5 s	3 s	4 s	5 s
Experienced drivers	0.86 (0.015)	0.92 (0.015)	0.90 (0.026)	0.39 (0.048)	0.43 (0.056)	0.55 (0.049)
Novice drivers	0.89 (0.008)	0.93 (0.011)	0.87 (0.022)	0.31 (0.045)	0.28 (0.039)	0.35 (0.050)

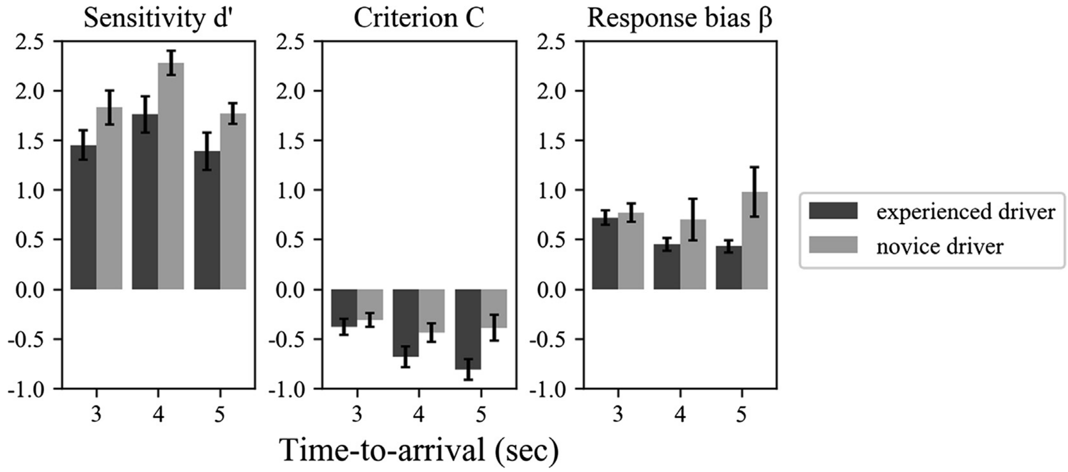

Fig. 3. Sensitivity (d'), criterion (C) and response bias (β) of drivers under different TTAs in intention judgments. Error bars represent standard errors.

76) = 4.19, $p = 0.019$, partial $\eta^2 = 0.099$). Further comparison showed that experienced drivers had a lower criterion for the larger TTA (i.e., 5 s) (all *ps* < 0.001), while novices kept a stable criterion for all TTAs (all *ps* > 0.07). As a result, the criteria of experienced drivers became significantly lower than novices when TTA reached 5 s ($p = 0.015$).

There was no significant main effect when response bias β was regarded as the dependent variable (F (1, 38) = 2.89, $p = 0.098$, partial $\eta^2 = 0.071$; F (2, 76) = 1.46, $p = 0.239$, partial $\eta^2 = 0.037$). The interaction effect was marginally significant, $F(2, 76) = 2.76$, $p = 0.069$, $\eta_p^2 = 0.068$. The test of simple effect showed that when the TTA was 5 s, the response bias β of experienced drivers was lower than novices ($p = 0.039$). In addition, the response bias β in all conditions were lower than 1, indicating a liberal criterion (tendency to say Yes, i.e., "the pedestrian will cross").

3.2.2. Reaction time (RT) of the intention judgment

The reaction time was defined as the time interval between the indicator presentation and participant's response. As the hypothesis, the intention recognition is time-consuming, so the differences in RT would reflect the process of intention recognition.

The means of reaction time over conditions are shown in Fig. 4. The trials with RT deviated beyond 2.5 times of standard deviation were excluded. A 2 (driving experience) × 3 (TTAs: 3, 4, 5 s) × 2 (intentions of pedestrians) mixed ANOVA was employed for the analysis of RT. Although experienced drivers made faster responses than novice drivers ($M = 762$ ms vs. $M = 846$ ms), RT was not significantly different across driving experience, $F < 1$. The main effect of TTAs was significant, F

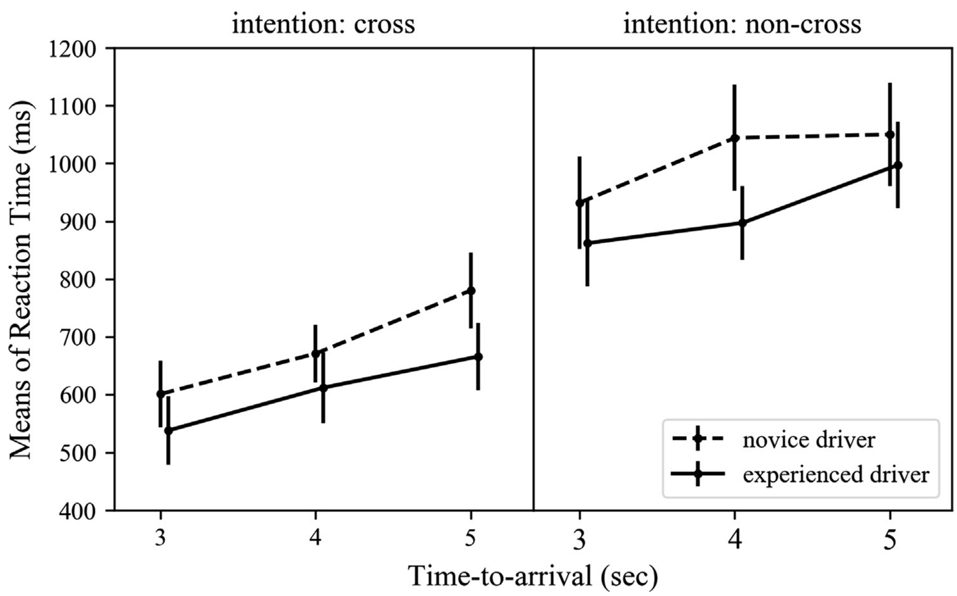

Fig. 4. Reaction time (RT) from a target pedestrian was indicated to response. Error bars represent standard errors.

(2, 76) = 33.50, $p < 0.001$, partial $\eta^2 = 0.469$. The pairwise comparisons revealed that the RT was longer for larger TTA (all *ps* < 0.001) when drivers had less time to recognize intentions. The effect of pedestrian intentions was also significant, with shorter RT in cross condition ($M = 645$ ms, $SE = 40$ ms) than non-cross condition ($M = 964$ ms, $SE = 54$ ms), $F(1, 38) = 114.42$, $p < 0.001$, partial $\eta^2 = 0.751$. None of the interaction effects was significant, all *ps* > 0.08.

Because participants did not get any feedback on their judgments, we also explored the response process according to their judgments ("will cross" or "will not cross"), rather than real intentions of pedestrians. A paired *t*-test showed that the RT was shorter for "will cross" judgments ($M = 738$ ms, $SE = 42$ ms) than "will not cross" judgments ($M = 915$ ms, $SE = 51$ ms), $t = -7.59$, $p < 0.001$, which means drivers were quicker in making a "will cross" judgment as compared with a "will not cross" judgment.

3.2.3. Eye movements during intention judgment

The following analyses concerned the fixation position and duration while processing pedestrian intentions. The eye movements during the last static frame (for a maximum of one second) were also included in the analyses, because even in the static frame, drivers could still get information on pedestrian intentions. Compared with the AOIs in the detection phase, the AOIs employed in this phase closely encompassed target pedestrians (see Fig. 5), allowing for more refined coding of eye movements. The fixation was defined as a gaze within the scene for at least 100 ms. The offset of a fixation, which meant the offset of the current fixation position relative to the center of the AOI, was calculated respectively in the horizontal and vertical direction. Then, the offset of a fixation position was transformed into a ratio scale ranged from 0 to 1 based on the width and height of the AOI (see Fig. 5).

Fig. 5 shows that drivers' fixations on the horizontal direction are symmetric around the center of pedestrians' body, with 48.3% of the fixations falling on a position smaller than 0.5. In contrast, in the vertical direction, more fixations fall on the upper body, with 66.4% of the fixations falling on the position smaller than 0.5.

When taking fixation duration into account, in order to carry on the statistical test, we simplified the fixation positions into three areas with the same size: top, middle, bottom zone. The mean fixation durations in the AOIs took into a 2 (driving experience) × 2 (intentions of pedestrian) × 3 (fixation positions: top, middle, bottom zone) mixed ANOVA. The results showed that the mean fixation duration was longer for experienced drivers ($M = 676$ ms, $SE = 27$ ms) than novice drivers ($M = 576$ ms, $SE = 27$ ms), $F(1, 38) = 6.65$, $p = 0.014$, partial $\eta^2 = 0.149$. When the mean of fixation duration was calculated with all fixations in the experiment (including eye movement in the normal driving section), there was no group difference ($M = 449$ ms for experienced drivers vs. $M = 433$ ms for novices), $t = 0.72$, $p = 0.477$. It reflected that the significant difference found in the ANOVA analysis was most likely due to the intention recognition task rather than other factors, excluding the possibility of general difference in normal driving style between experienced and novice drivers.

The main effect of pedestrian intentions was also found, with longer fixation duration for pedestrians intended to cross ($M = 661$ ms, $SE = 20$ ms) than those did not intend to cross ($M = 592$ ms, $SE = 21$ ms), $F(1, 38) = 27.44$, $p < 0.001$, partial $\eta^2 = 0.419$. The fixation duration also varied with positions, $F(2, 76) = 16.88$, $p < 0.001$, partial $\eta^2 = 0.308$, which was longer for upper than lower zone (see Fig. 6, all *ps* < 0.005. No interaction effect was found, all *ps* > 0.3.

3.3. Detection and intention recognition as a continual process

Will drivers' speed of intention recognition be affected by detection performance? For trials coded as "detected before indicator" as mentioned, their RTs ($M = 776$ ms, $SE = 43$ ms) were significantly shorter than the RTs ($M = 1157$ ms,

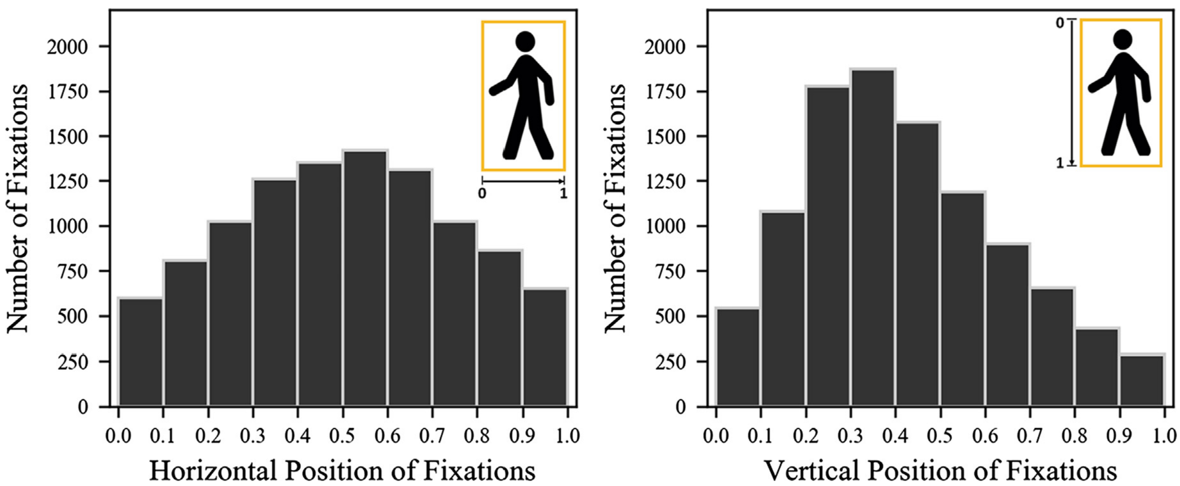

Fig. 5. The distribution of fixation across the horizontal direction (left) and the vertical direction (right). The definition of fixation positions is shown at the top-right of each figure.

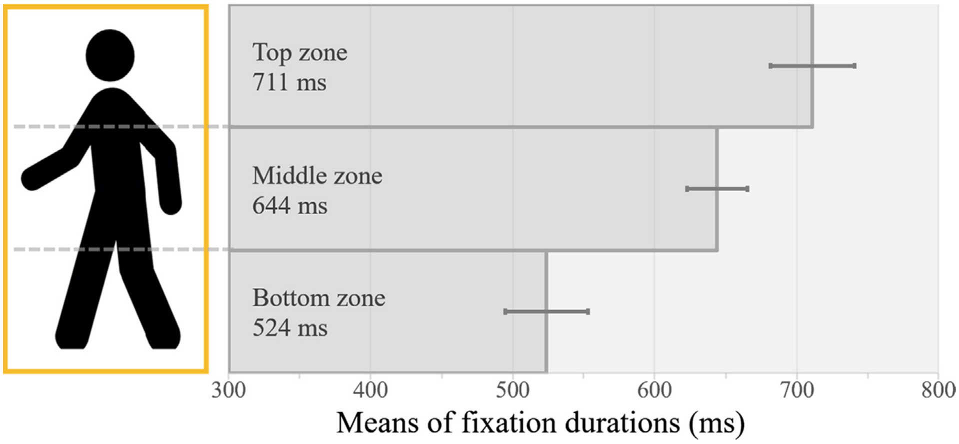

Fig. 6. The means of fixation duration on the different parts of a target pedestrian. Error bars represent the standard errors.

SE = 61 ms) of trials coded as "not detected before indicator" ($t = -12.49$, $p < 0.001$). A possible reason is that if the drivers directed their attention to pedestrians before indicator, their intention recognition process would begin earlier, resulting in shorter RTs. To verify the possibility and exclude conditions where participants turned to other things after detection of pedestrians, we conducted a Spearman's correlation (two-tailed) to estimate the relationship between dwell time (the total of fixation time in the AOI in one trial) and RT for trials coded as "detected before indicator". The significant correlation ($r = -0.208$, $p < 0.001$) indicated that the RT was shorter for longer dwell time (longer processing duration). To sum up, if drivers detected target pedestrians earlier ("detected before indicator") and kept monitoring for a longer time (dwell time), they would make a quicker response to pedestrian intentions.

4. Discussion

The current study explored the drivers' intention recognition process from detection to judgments when interacting with pedestrians. In the detection phase, both experienced and novice drivers had a higher detection rate, quicker detection and longer first fixation duration toward pedestrians who intended to cross than those did not intend to cross. As to the performance of intention judgments, experienced drivers were less accurate than novice drivers, with lower sensitivity. However, they had more flexible criterion corresponding to the change of TTAs. Pedestrians intending to cross also attracted a longer mean fixation duration, especially for experienced drivers. Within a specific pedestrian, the upper body attracted more and longer fixations than the lower part. Overall, the detection and intention recognition phase both affected final responses. The response was quicker when TTA was shorter, pedestrians intended to cross or drivers paid more attention to pedestrians. The implications of these findings are discussed in this section.

4.1. The accuracy and bias of the intention judgment

As to the accuracy of judgment, drivers with different experience failed to recognize the intentions of pedestrians intended to cross in about 8% of the 109 trials. Although the miss rate was apparently low, it was worth noting in terms of safety. If drivers made a wrong judgment to whom intended to cross, a conflict would happen. In contrast, even if drivers had a false alarm, it would not cause any conflict. Therefore, the traditional evaluation of performance with accuracy is not appropriate on this occasion. Instead, we evaluated drivers' ability to distinguish pedestrians who intended to cross from those did not intend to cross. In regard to group differences between experienced drivers and novice drivers, plenty of studies found that experienced drivers usually outperformed novice drivers in detection rate, judgment accuracy, processing efficiency and situation awareness in hazard perception tasks (Borowsky, Shinar, & Oron-Gilad, 2010; Crundall, 2016; Crundall et al., 2012; Jackson, Chapman, & Crundall, 2009; Ventsislavova et al., 2016). However, in the current study, the result revealed the apparently better performance of novice drivers with higher sensitivity. One possible reason is that our study focused on the processing of intentions, and the target pedestrians were easy to notice, whereas, in hazard perception, drivers need to predict whether potential hazards would appear based on their experience.

The lower sensitivity of experienced drivers did not mean riskier judgments, as experienced drivers showed more conservative and flexible strategies than novice drivers. The criteria of experienced drivers were relatively lower than novices' and changed over TTAs. More specifically, experienced drivers were more conservative at larger TTA (tending to report "will cross"), while novice drivers kept a stable criterion regardless of the changes in TTA. As a result, their difference in criterion became larger with increasing TTAs (usually means farther distance) and finally became significant when TTA reached 5 s.

The higher tendency to report "will cross" of experienced drivers may reflect a lower level of risk acceptance (Deery, 1999). A similar phenomenon was also found in a hazard perception test: experienced drivers preferred to make an evasive maneuver than novices, even for the quasi-hazardous situations (Ventsislavova et al., 2016).

The changing criterion relative to TTAs also indicates that experienced drivers may have a better prediction of pedestrian behaviors than novice drivers. According to studies on pedestrians' gap acceptance and violation behaviors, pedestrians are more likely to cross at larger TTAs (Brewer, Fitzpatrick, Whitacre, & Lord, 2006; Wang, Wu, Zheng, & McDonald, 2010) and they may also run across the road instead of crossing in a constant speed (Zhuang and Wu, 2012, 2011). It was possible that experienced drivers considered the variability of pedestrians' behaviors on the road, and predicted pedestrians' behaviors based on their theory of mind about pedestrians. A piece of indirect evidence for this hypothesis was that drivers interacted with human pedestrians and robot pedestrians in different ways (Lehsing, Benz, & Bengler, 2016). In another study conducted by Lee (2016), the researcher measured drivers' mentalizing skills, in addition to the main task to judge the intentions of road users. The result demonstrated a trend towards a significant positive correlation between drivers' mentalizing skills and sensitivity in intention judgments of other road users.

4.2. Process and mechanism in intention recognition

In normal driving, an appropriate response to interacting pedestrians requires successful detection, correct recognition, and timely response. In this study, the focus is the intention recognition phase, but the findings showed that previous processes affected the final responses. Compared with the pedestrians not been detected before the presentation of rectangular indicator (pointing out the target pedestrian), those who were detected before the indicator resulted in faster responses. For them, the longer drivers gazed at (i.e., dwell time), the faster responses were. Therefore, findings from all phases need to be integrated to probe into the cognitive mechanism of intention recognition.

In the detection phase, all drivers had a higher detection rate as well as shorter latency of detection towards pedestrians with "cross" intention than those with "non-cross" intention. The easier detection might be caused by salient visual features like close standing position, high walking speed or other aggressive behaviors that are often seen in pedestrians intending to cross but not available in those did not intend to cross. These behaviors also explained the longer first fixation duration towards pedestrians with "cross" intention, as hazardous behaviors could induce an attentional capture effect (Underwood, 2007; Underwood, Phelps, Wright, Van Loon, & Galpin, 2005). From the drivers' aspect, the longer first fixation duration may indicate processing of information related to pedestrian intentions at the moment of detection. Although we named this process as the detection phase, drivers were in fact continuously evaluating the risk levels of interacting objects. In natural driving scenarios as well as in this study, there were usually multiple objects and pedestrians simultaneously visible in a scene. Drivers need to make a rough assessment of them to focus on relatively more hazardous targets. In Tsimhoni, Kandt, and Flannagan's study (2008), drivers were asked to evaluate the necessity of a pedestrian needed to be monitored and they found that pedestrian's position and direction influenced drivers' ratings. Therefore, it is possible that for pedestrians would not cross, drivers simply glanced at them and tagged them as safe. For those who intended to cross, drivers needed to interpret their intentions to decide whether further monitoring was needed. Accordingly, we suppose that the longer first fixation duration may reflect an initial evaluation of intentions to screen the potential hazards before an object is determined for further elaborate intention judgments.

In the intention recognition phase, all drivers had longer mean fixation duration on the pedestrians who intended to cross. In previous studies, longer fixation duration meant that gazes were fixed on task-relevant objects or locations (Lappi, Rinkkala, & Pekkanen, 2017). Fixation duration was also used to reflect the depth of processing of hazards in many studies (Chapman, Underwood, & Roberts, 2002; Chapman & Underwood, 1998; Kovácsová et al., 2018; Velichkovsky, Rothert, Kopf, Dornhöfer, & Joos, 2002). Therefore, the longer fixation duration may reflect more attention allocated to pedestrians intending to cross and a higher level of information processing of their intentions.

However, the question remains as to what information drivers utilized to recognize intentions and how that led to final judgments. The common process of choice and judgment beyond intention judgment could be described with the dynamic signal detection theory (Neal & Kwantes, 2009; Pleskac & Busemeyer, 2010). In this theory, participants collect information from many sources and transform them as pieces of evidence to support potential judgments. If evidence accumulated for any judgment reaches a threshold, the evidence accumulation will stop, and the judgment will be made. Adapting this theory to the drivers' judgments of pedestrian intentions, it was possible that drivers collected information from pedestrians' behaviors, road contexts, and memories to support two potential judgments: "will cross" and "will not cross". Since the general principle in traffic context is to ensure safety, the threshold for making a "will cross" judgment should be much lower than "will not cross" judgment. If this is indeed the case, we would observe several phenomena: (1) Drivers will pay more attention to the area that provides richer information on crossing intentions to gain evidence. (2) Drivers will have a quicker response if more evidence is accumulated. (3) The responses of "will cross" judgments will be faster than "will not cross" judgments.

All these predictions were confirmed. First, the analysis of fixation positions showed that drivers tended to spend a longer time to gaze at the upper part of pedestrians. Although the current experiment was not able to distinguish exactly which body part they are looking at, the upper 1/3 implied the head part. In Schmidt et al.'s study (2008), head activity was reported as the most important cue to pedestrian intentions. Similarly, in the driver-cyclist interaction, face, eye-contact, and head movement were proved to be important social signals of cyclist intentions (Walker, 2005; Walker & Brosnan, 2007). Therefore, drivers did pay more attention to the information-rich area. Second, corresponding to the prediction (2), drivers did respond faster when more evidence had been accumulated. As mentioned above, for pedestrians detected earlier and gained longer dwell time, the responses were quicker. Similarly, we observed shorter RT for pedestrians at smaller TTA, which meant a longer time to accumulate evidence before intention judgments. Third, corresponding to the prediction (3), we found that both groups of drivers tended to make "will cross" judgments, and the responses were faster than "will not cross" judgments. One reason might be the lower threshold of "will cross" judgments to ensure safety. Considering that most "will cross" judgments were made when pedestrians did have the intention to cross (hit rate was 0.92 for experienced drivers, 0.91 for novice drivers), it was also possible that pedestrians intending to cross had more obvious communication cues to be transformed as evidence supporting "will cross" judgments. In contrast, for the "non-cross" condition, fewer cues were available to make a judgment. The final "will not cross" judgment might be a consequence of "no evidence for *will cross* was found". In fact, drivers do have slower responses (up to 319 ms) if pedestrians did not intend to cross than if they intended to cross.

Combining the findings in all phases, we propose that intention recognition may consist of two phases: a rough evaluation closely after detection to screen potential hazards and an elaborate intention recognition in agreement with an evidence accumulation process. However, as a primitive study on intention recognition during the driver-pedestrian interaction, the current study could only provide some indirect evidence to support this postulation. Further studies are needed to discriminate the two phases and dive into the mechanism in each phase.

4.3. Implications and further research

In addition to the theoretical questions raised above, our findings have several additional implications for future research. First, we found that the time available to process intentions was crucial to the speed of responding. Therefore, future studies on intention recognition need to take the time dimension into account. Second, currently, autonomous driving is gaining ground, but its success depends on safe interaction with pedestrians, which involves correct recognition of pedestrian intentions. Our finding showed that human drivers relied more on the upper body to recognize intentions, rather than the body bending and spread of legs which used by some intention recognition algorithms (Köhler et al., 2012).

Future studies can also compare the performance of machine drivers with human drivers in comparable contexts. If the former is weaker, then we can add or provide more weights on cues from the upper body to improve the algorithm. On the other way around, if computer vision exceeded the performance of human drivers, then we can probably train human drivers with the cues found by the algorithm or facilitate their judgments with ADAS to reduce uncertainty (Houtenbos, Jagtman, Hagenzieker, Wieringa, & Hale, 2005). This is especially important given that drivers tend to delay their responses when cues of intentions are inadequate.

4.4. Limitations

The current study simulated the driver-pedestrian interaction situation with videos in the laboratory. The task was to make an intention judgment with a joystick without the built-in requirement to control the vehicle. As a result, the workload of the driver was lower in the experiment than in real traffic. It probably benefited drivers' performance, especially novice drivers who were younger and more familiar with the joystick as an input device. Therefore, although we found a shorter yet not statistically significant RT for experienced drivers than novices (762 ms vs. 846 ms), it was possible that experienced drivers might take actions faster than novices in real driving scenes. A potential way to overcome this limitation is to study intention judgments in a more ecological context. Another related limitation was that pedestrians' actual crossing behaviors observed afterward were used as the ground truths of their intentions, but their real intentions might also depend on drivers' responses and the overall traffic condition. In real traffic context, the driver-pedestrian interaction is bi-directional. The best way to study the interaction process is to involve two parties at the same time. However, real interaction behaviors and underlying information processing are hard to capture in natural scenes, and simulators are far from satisfactory. Currently, most driving simulators only gives a built-in module of pedestrians that can simulate very simple and rigid behaviors rather than take real pedestrian behaviors as input. At the same time, pedestrian simulators only afford a built-in module of vehicle behaviors rather than read real driver behaviors as input. To determine the detailed process of interaction, the future simulator should be able to combine the two and allow simultaneous simulation of natural interaction behaviors within the same scene to promote development in this field (Lehsing et al., 2016).

5. Conclusions

Making a judgment of pedestrian road crossing intentions requires detection of pedestrians and recognition of their intention. In the detection phase, an attentional capture effect was found for pedestrians who intended to cross: Drivers detected them earlier and gazed for a longer duration. During intention recognition, eye movements indicated that experienced drivers took a higher level of information processing than novices. However, both experienced drivers and novice drivers paid more attention to the information-rich area (the upper body of pedestrians) for a longer duration to gain evidence for intention judgments. The final judgments of crossing intentions were conservative (prefer a "will cross" judgment), especially for experienced drivers and pedestrians at a greater time-to-arrival (TTA). Besides, the responses of intention judgments were much slower when the TTA was greater or when pedestrians did not intend to cross.

Overall, these findings show that the intention recognition may include two phases: one closely following the detection phase to roughly screen potential hazards, and one elaborate evaluation of intentions that can be explained by the evidence accumulate process.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.trf.2019.07.004.

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